

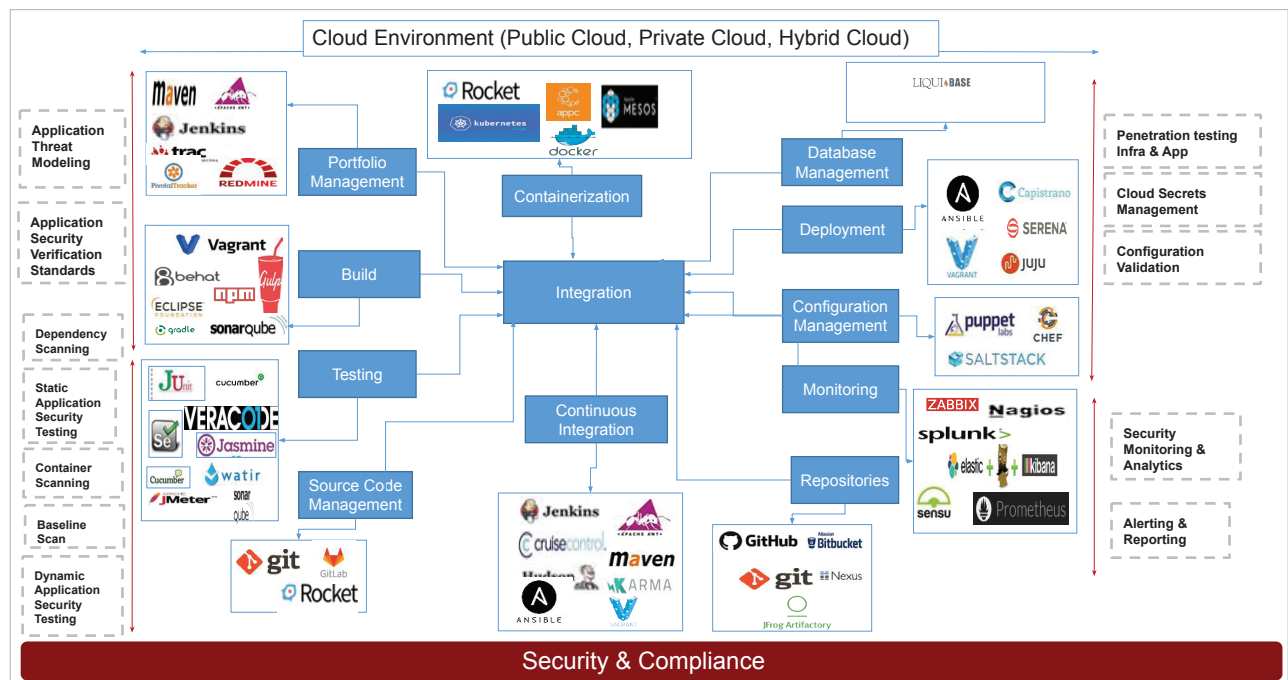


Figure 1: DevSecOps life cycle and mapping of open source tools

Git and GitLab are the leading source control systems. Git allows developers to collaborate with each other on a distributed version control system. GitLab provides the centralised and integrated platform for developers.

Testing: Continuous testing promotes organisation-wide cultural change to promote capabilities like test early, test faster and automate. Continuous testing synchronises testing and QA with Dev and Ops processes that are optimised to achieve business and development goals.

Tools like Tosca, Selenium, Veracode, SonarQube, Cucumber and JUnit are used to automate the execution of test cases.

Continuous integration: Continuous integration (CI) helps developers to integrate code into a shared repository several times a day. It allows teams to detect problems early and verifies each check-in. By integrating regularly, it can detect errors quickly and locate them more easily.

The most popular CI tool in the market is Jenkins. Other popular CI tools are Bamboo and Hudson.

Continuous deployment: In continuous deployment, every change goes through the pipeline and is automatically put into production, resulting in many production deployments every day with greater delivery speed and frequency for complex applications.

Ansible, Kamatera and Vagrant are the most useful continuous deployment tools used in the cloud environment.

Configuration/provision management: Configuration management helps in establishing and maintaining consistency in an application's functional requirements and performance. Configuration management tools work based on the master-slave architecture.

Popular configuration management tools used in the cloud environment are Puppet, Chef, Ansible, and SaltStack.

Containerisation: Containerisation tools help in maintaining consistency across the environments where the application is developed, tested and deployed. Containerisation eliminates the failure in a production environment by packaging and replicating the same dependencies and packages that are used in the development, testing and staging environments.

Docker is the most used containerisation tool.

Repositories: An artifact repository is a collection of binary software artifacts and metadata stored in a defined directory structure. A repository stores two types of artifacts — releases and snapshots. Release repositories are for stable, static release artifacts, and snapshot repositories are frequently updated repositories that store binary software artifacts from projects that are under constant development.